



CONTACT

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PROFILE

M.Sc. in Computer Science at TU Dresden with a focus on Human-Computer Interaction, accessibility, and intelligent interactive systems. Experienced in developing Python-based research prototypes, multimodal interfaces, and user-centered evaluations, especially for blind and visually impaired users.

Available to start immediately

SKILLS

Programming Languages
Python, Java, C++

Backend
Spring Boot, WebSocket, SQL

Frontend
Vue.js, HTML/CSS

Computer Vision
OpenCV, MediaPipe

Tools & Libraries
Git, Docker, NumPy

LANGUAGES

English - C1
German - C1
Chinese - Native

EXPERIENCE

Research Assistant 04/2026 – Present
TU Dresden, Chair of Measurement and Sensor System Technique

- Development of a visualization interface for a training system of optical diffractive neural networks (ODNN)
- Design and implementation of a web-based visualization interface using Streamlit to support parameter configuration, training monitoring, and result analysis

Research Assistant 02/2026 – 03/2026
TU Dresden, Chair of Human-Computer Interaction

- Optimized and further developed the master thesis prototype.
- Planned and conducted a user study in cooperation with a project partner in Nuremberg (participants: blind students)
- Published the results as a conference paper at ICCHP [\[Paper\]](#)

MultiSense Dashboard — Accessible Real-Time Monitoring 10/2024 – 04/2025
TU Dresden

- Developed a web-based dashboard for real-time monitoring of multimodal sensor data for blind and visually impaired users.
- Primarily responsible for frontend development, configuration logic, and dashboard visualization using Vue.js.
- Integrated real-time data from backend services (Apache Kafka/WebSocket) and implemented accessible interaction components (speech/sonification).

Smart ANC — Adaptive Noise Masking 04/2024 – 09/2024
TU Dresden

- Contributed to a subproject of the AutARK research project on adaptive masking of disturbing environmental noise in office contexts.
- Primarily responsible for the module for masking strategy and the real-time generation and adaptation of masking sounds.
- Implemented signal- and data-driven logic in Python to respond to different types of noise.

EDUCATION

M.Sc. in Computer Science 04/2025 – 03/2026
TU Dresden

Studied in Computer Science at TU Dresden since 10/2022; initially enrolled in the Diplom degree program, and since 04/2025 in the M.Sc. Computer Science program with full recognition of previously completed coursework.

Master Thesis: Multimodal Spatial Arithmetic with Tangibles

- Developed a multimodal learning and interaction system for blind and visually impaired users to support spatial arithmetic using a tactile grid and tangibles.
- Designed and implemented a Python prototype including grid recognition, tangible detection, hand tracking, and arithmetic logic.
- Integrated camera-based image recognition on Raspberry Pi as well as speech processing (ASR/TTS) for guidance and feedback.
- Final Grade: 1.9

B.Sc. Computer Science and Technology 09/2017 – 07/2021
Shanghai University of Electric Power

- Bachelor's thesis: Development of a C++ program on a microcontroller for implementing the RTCP algorithm.
- Final grade: 1,7